

Post Composer Mini-Project Setup Guide

This guide will walk you through the process of setting up and running the Post Composer mini-project, which consists of a React frontend and a Spring Boot backend.

Prerequisites

Before you begin, ensure you have the following installed on your machine:

- **Java 17** (or higher)
- **Node.js** (v14 or higher) and **npm**

1. Running the Backend (Spring Boot)

The backend is built with Spring Boot and uses an in-memory H2 database, which means you don't need to install or configure any external database like MySQL.

1. Open a terminal.
2. Navigate to the backend directory inside the project folder:

```
cd exp_5_code/backend
```

3. Run the application using the included Gradle wrapper:

- On **Mac/Linux**:

```
./gradlew bootRun
```

- On **Windows**:

```
gradlew.bat bootRun
```

4. The backend server will start on <http://localhost:8080>. Keep this terminal window open.

2. Running the Frontend (React)

The frontend is built with React and Vite.

1. Open a **new** terminal window (leave the backend terminal running).
2. Navigate to the frontend directory:

```
cd exp_5_code/frontend
```

3. Install the required Node dependencies (you only need to do this once):

```
npm install
```

4. Start the development server:

```
npm run dev
```

5. The frontend application will be available at <http://localhost:5173>. Open this URL in your web

browser.

Testing the Application

Once both servers are running:

1. Go to `http://localhost:5173`.
2. You will see the Post Composer interface.
3. Select a platform (Twitter, Instagram, or Facebook).
4. Start typing your post. Notice the word count updating at the bottom.
5. If you exceed the word limit for the selected platform, a red global error banner will appear at the top, and you won't be able to add more words.
6. Click **Post** to save your message. It will appear in the "Recent Posts" list below, where you can easily edit or delete it.

Troubleshooting

- **Port in use:** If port 8080 or 5173 is already in use by another application, you will need to stop that application first.
- **Dependencies not found:** Ensure you ran `npm install` in the frontend directory before running `npm run dev`.